

A Pitfall with Estimated DSGE-Based Government Spending Multipliers[†]

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This paper examines issues related to the estimation of the government spending multiplier (GSM) in a DSGE context. We stress a source of bias in the GSM arising from the combination of endogenous government expenditures and Edgeworth complementarity between private consumption and government expenditures. Due to cross-equation restrictions, omitting the endogenous component of government policy at the estimation stage would lead an econometrician to underestimate the degree of Edgeworth complementarity and, consequently, the long-run GSM. An estimated version of our model with US postwar data shows that this bias matters quantitatively. The results are robust to a number of perturbations. (JEL E13, E23, E32, E62, H50)

Standard practice in estimation of dynamic stochastic general equilibrium (DSGE) models (e.g., the well-known work by Smets and Wouters 2007) is to assume that government consumption expenditures are described by an exogenous stochastic process and are separable in the households' period utility function. This standard practice has been adopted in the most recent analyses of fiscal policy (e.g., Christiano, Eichenbaum, and Rebelo 2011; Coenen et al. 2012; Cogan et al. 2010; Drautzburg and Uhlig 2011; Eggertsson 2011; Erceg and Lindé 2010; Fernández-Villaverde 2010; and Uhlig 2010).

In this paper, we argue that both short-run and long-run government spending multipliers (GSM) obtained in this literature may be downward biased. This is so because the standard approach does not typically allow for the possibility that

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private consumption and government spending are *Edgeworth complements* in the utility function¹ and that government spending has an *endogenous countercyclical component* (automatic stabilizer) (e.g., Cúrdia and Reis 2010; Jones 2002; Leeper, Plante and Traum 2010; and McGrattan 1994).

A countercyclical component in the government spending rule and Edgeworth complementarity work in opposite directions in terms of generating a positive correlation between output and government expenditures. Intuitively, Edgeworth complementarity tends to increase the correlation between output and government spending, since under such a configuration an increase in government expenditures would induce people to consume more. At the same time, a countercyclical component in the policy rule reduces this correlation, by construction. This yields a trade-off. Given an observed pattern of correlation between output and government spending that an analyst seeks to match, a highly countercyclical endogenous component of policy must be compensated by a high degree of Edgeworth complementarity. Since, as we show, the GSM increases with the degree of Edgeworth complementarity, this will mechanically translate into a large GSM. Conversely, omitting the endogenous component of policy will imply a small degree of complementarity, thus yielding a smaller GSM.

To establish these results, we assess the potential quantitative bias of standard practice in a stylized model that allows us to derive analytical results for both short-run and long-run GSMs and the empirical relevance of these channels in an estimated neoclassical growth model and examine the robustness of these empirical results in a benchmark New Keynesian model.

Importantly, we find that the standard approach may mask sizable crowding in effects on private consumption, as the standard no automatic stabilizer assumption results in a low degree of Edgeworth complementarity. If automatic stabilizers are properly accounted for, then a higher degree of Edgeworth complementarity is needed to match the correlation between private consumption and government spending. As the GSM increases with the degree of Edgeworth complementarity, the standard empirical approach may also result in a downward-biased estimate of the GSM.

In our benchmark empirical specification with Edgeworth complementarity and a countercyclical component of policy, the estimated long-run multiplier amounts to 1.31. Using the same model and imposing an exogenous policy rule, we obtain a multiplier of 0.97, significantly smaller than our benchmark value by 0.34 points. This difference is even bigger when both Edgeworth complementarity and the countercyclical component of policy are omitted, in which case the estimated multiplier is approximately equal to 0.5. Such a difference is clearly not neutral if the model is used to assess recovery plans of the same size as those recently enacted in the United States. To illustrate this more concretely, we feed the American Recovery and Reinvestment Act (ARRA) fiscal stimulus package into our model. We obtain that omitting the endogenous policy

¹ We say that private consumption and government spending are Edgeworth complements/substitutes when an increase in government spending raises/diminishes the marginal utility of private consumption. Such a specification has now become standard, following the seminal work by Aschauer (1985), Bailey (1971), Barro (1981), Braun (1994), Christiano and Eichenbaum (1992), Finn (1998), and McGrattan (1994).

rule at the estimation stage would lead an analyst to underestimate the short-run GSM by slightly more than 0.25 points. Clearly, these are not negligible figures.

To complement these results, we consider several robustness analyses. First, we investigate the robustness of policy countercyclicality. To do so, we reestimate our DSGE model under a large battery of alternative specifications for the endogenous component of government spending. The bottom line is that our benchmark specification is supported by the data. Second, to address potential issues of policy rules instability, we reestimate the model over two subsamples. Although the estimated parameters show moderate signs of time-variability, our main conclusion still holds. Finally, as mentioned above, we investigate whether our results still obtain in a Smets-Wouters type model. Although this model yields different multipliers than in our benchmark specification, we still find that omitting the endogenous component of government expenditures would severely downward-bias the estimated multiplier.

Our estimations yield a GSM that exceeds unity and a near zero multiplier for private consumption. These findings are broadly in the range of values reported by Hall (2009)—typically in between 0.5 and 1.7 for output. This range of values derives from various methods. These include, for example, GSM estimates from DSGE models (e.g., Christiano, Eichenbaum, and Rebelo 2011; Monacelli, Perotti, and Trigari 2010; and Zubairy 2010), single regressions on government purchases (Barro and Redlick 2009, and Hall 2009), and Structural Vector Autoregressions (SVARs) (e.g., Blanchard and Perotti 2002; Caldara and Kamps 2008; and Fisher and Peters 2010). Interestingly, the DSGE literature itself generates such a range of values for the GSM: the lower bound obtains in typical calibrated neoclassical setups; the upper bound is obtained in Christiano, Eichenbaum, and Rebelo (2011). A key contribution of the latter is to show under which circumstances the multiplier can be much larger than one, typically when the economy has reached the zero lower bound on the nominal interest rate. Our paper adds to this literature by spotting possible estimation biases that naturally emerge from such models. The strength of these models lies precisely in their cross-equation restrictions, which significantly contribute to highlight identification problems for the GSM.²

The rest of the paper is organized as follows. In Section I, we expound the simple model and illustrate the trade-off between Edgeworth complementarity and the countercyclical component of policy in terms of matching the observed correlation between output and government expenditures. We then characterize the bias that would result from omitting this countercyclical component. Section II develops a quantitative version of this model that we take to postwar US data. We then explore the quantitative implications of policy rule misspecification. In Section III, we investigate the robustness of our results. The last section briefly concludes.

I. A Simple Illustrative Example

In this section, we work out an equilibrium model simple enough to obtain closed-form formulas illustrating how short- and long-run government spending multipliers are

² Our analytical results also point out to potential sources of bias in SVARs, as emphasized by Caldara (2011).

biased when the econometrician omits the endogenous component of public policy. We focus on Edgeworth complementarity/substitutability between private consumption and public spending as the transmission mechanism of government expenditures. The literature has considered other mechanisms, such as nonseparable utility, externalities on preferences and technology, or deep habits. Whatever the mechanism considered, the log-linear equilibrium output would take the same form as that obtained below.³

A. The Model

Consider a discrete time economy populated with a large number of infinitely-lived, identical agents. The representative household seeks to maximize

$$(1) \quad E_t \sum_{i=0}^{\infty} \beta^i \left\{ \log(c_{t+i} + \alpha_g g_{t+i}) - \frac{\eta}{1+\nu} n_{t+i}^{1+\nu} \right\}$$

subject to the sequence of budget constraints ($t \geq 0$)

$$(2) \quad c_t + E_t \{q_{t,t+1} b_{t+1}\} = w_t n_t - T_t + b_t,$$

where $E_t\{\cdot\}$ is the expectation operator, conditioned on information available as of time t ; $\beta \in (0, 1)$ is the subjective discount factor; c_t is private consumption; g_t denotes public expenditures; n_t is the labor supply; w_t is the real wage rate; and T_t denotes lump-sum taxes. The Frisch elasticity of labor supply is $1/\nu$ and $\eta > 0$ is a scale parameter. Finally, b_{t+1} represents the value in period $t + 1$ of the household's bond portfolio at the end of period t . As we assume complete financial markets, this portfolio includes state-contingent claims. The market value of such a bundle of state-contingent claims in period t is given by $E_t\{q_{t,t+1} b_{t+1}\}$ where the random variable $q_{t,t+1}$ is the stochastic discount factor for pricing arbitrary financial claims.

The parameter α_g , in turn, accounts for the complementarity/substitutability between private consumption c_t and public spending g_t .⁴ If $\alpha_g \geq 0$, government spending substitutes for private consumption, with perfect substitution if $\alpha_g = 1$, as in Christiano and Eichenbaum (1992). In this case, a permanent increase in government spending has no effect on output and hours but reduces private consumption, through a perfect crowding-out effect. In the special case $\alpha_g = 0$, we recover the standard business cycle model, with government spending operating through a negative income effect on labor supply (see Aiyagari, Christiano, and Eichenbaum 1992; and Baxter and King 1993). When the parameter $\alpha_g < 0$, government spending complements private consumption. Then, it can be the case (depending on the labor supply elasticity) that private consumption will react positively to an unexpected increase in government spending.

³ See the online Appendix for a review of alternative transmission mechanisms of government spending.

⁴ Here we use the specification adopted by Christiano and Eichenbaum (1992); Finn (1998); McGrattan (1994); among others. Alternatively, CES specifications of utility have been considered (see Bouakez and Rebei 2007; McGrattan, Rogerson, and Wright 1997). These yield the exact same log-linearized equilibrium conditions as our specification.

The representative firm produces a homogeneous final good y_t using labor as the sole input, according to the constant returns-to-scale technology

$$y_t = e^{z_t} n_t.$$

Here, z_t is a shock to total factor productivity, assumed to be *independently and identically distributed* with $z_t \sim N(0, \sigma_z^2)$. Profit maximization implies that the marginal productivity of labor equals the real wage, that is $w_t = e^{z_t}$.

Government purchases are entirely financed by taxes,

$$T_t = g_t.$$

As in the literature emphasizing the relevance of stabilizing government spending rules (see, among others, Cúrdia and Reis 2010; Jones 2002; Leeper, Plante, and Traum 2010; and McGrattan 1994), we specify a feedback rule of the following form:

$$(3) \quad g_t = \bar{g} \left(\frac{y_t}{y_{t-1}} \right)^{-\varphi_g} e^{u_t}, \quad \varphi_g \geq 0,$$

where $\bar{g}$ is a scale factor that pins down the deterministic steady-state level of government expenditures and φ_g governs the responsiveness of g_t to output growth. The random term u_t represents the discretionary part of policy and is assumed to be *independently and identically distributed* with $u_t \sim N(0, \sigma_u^2)$. A simpler rule would have government spending react to current output only. A problem with such a specification within our simplified model is that it would compromise identification of the policy parameter.⁵ In anticipation of the next section, we also notice that the dynamic rule (3) is favored by the data when we estimate a quantitative model version.

Finally, the market clearing condition on the goods market writes

$$y_t = c_t + g_t.$$

Combining the household's first-order condition on labor, the profit maximization condition, and the resource constraint, one finally arrives at the equilibrium condition

$$(4) \quad \eta y_t^\nu = \frac{e^{(1+\nu)z_t}}{y_t - (1 - \alpha_g)g_t}.$$

Conditions (3) and (4) together constitute the equilibrium system governing the dynamics of the above economy. To ensure positiveness of the steady-state marginal utility of consumption, we henceforth impose the restriction $\alpha_g > (s_g - 1)/s_g$, where $s_g \equiv \bar{g}/\bar{y} \in [0, 1)$ is the steady-state public spending-output ratio, $\bar{y}$ being the steady-state level of output. Also, we rule out more than perfect substitutability between private consumption and public expenditures by imposing $\alpha_g \leq 1$.

⁵ Under such a simple rule with contemporaneous output, the reduced-form involves three free parameters while the structural model features four free parameters.

In this economy, the long-run GSM is defined as follows.

DEFINITION 1: *The long-run government spending multiplier, denoted by $\Delta y/\Delta g$, is the increase in steady-state output $\bar{y}$ consecutive to an increase in steady-state government spending expenditures $\bar{g}$*

$$\frac{\Delta y}{\Delta g} \equiv \frac{d\bar{y}}{d\bar{g}}.$$

From this definition and the structure of the above model economy, the following proposition states key properties of the long-run GSM.

PROPOSITION 1: *Under the preceding assumptions:*

(i) *The long-run government spending multiplier $\Delta y/\Delta g$ is*

$$\frac{\Delta y}{\Delta g} = \frac{1 - \alpha_g}{1 + \nu[1 - s_g(1 - \alpha_g)]}.$$

(ii) *The multiplier is a decreasing function of α_g .*

PROOF:

See Appendix.

This proposition establishes that the long-run GSM depends on the share of government spending in output (s_g), on the inverse of the Frisch elasticity of labor (ν), and on the parameter governing the degree of Edgeworth complementarity between private consumption and government expenditures (α_g). Notice that given the parameter restrictions discussed above, the multiplier is in $[0, 1/s_g]$.

The proposition establishes that the multiplier is a decreasing function of α_g . Indeed, as α_g increases, private consumption and public expenditures become more and more substitutable. Thus, the crowding out effect of government consumption mechanically implies a lower multiplier. Notice also that the multiplier is a decreasing function of ν . Indeed, the higher ν , the lower the labor supply elasticity. In turn, a lower labor supply elasticity mutes the negative wealth effect of government spending, thus resulting in a lower multiplier. Finally, the multiplier is an increasing function of the share of government spending in output s_g . Given an elasticity ν , this parameter governs the overall size of the negative wealth effect of government spending. Thus, a large s_g results in a large multiplier.

For later reference, it is useful to define

$$\alpha \equiv s_g \frac{\Delta y}{\Delta g}.$$

Given the above parameter restrictions, it must be the case that $\alpha \in [0, 1]$.

B. A Trade-Off between Two Conflicting Mechanisms

We now present an argument that the GSM, or equivalently α , might be biased at the estimation stage. While in our setup, α does not depend directly on the degree of countercyclicality of the endogenous component of government policy (φ_g), the main thesis of this paper is that φ_g can contaminate the long-run GSM indirectly. This is the outcome of the interplay between two conflicting economic forces: Edgeworth complementarity between private consumption and government spending on the one side and the countercyclicality of the endogenous component of government spending on the other side.

To gain intuition as to how this can happen, we start by loglinearizing the system (3)–(4) in the neighborhood of the deterministic steady state. This yields

$$(5) \quad \hat{y}_t = \alpha \hat{g}_t + \xi z_t$$

$$(6) \quad \hat{g}_t = -\varphi_g(\hat{y}_t - \hat{y}_{t-1}) + u_t,$$

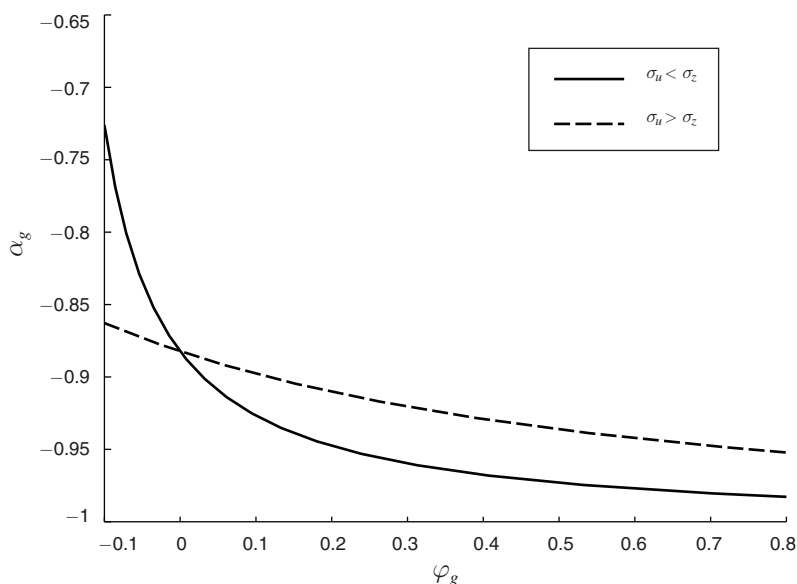
where a letter with a hat denotes the log deviation (with respect to steady-state value) of the associated variable. Here, ξ is a composite parameter. In the remainder, to simplify the algebra, we drop this coefficient from the dynamic system. This can always be done by rescaling appropriately the standard error of z_t .

The system (5)–(6) makes clear how the degree of countercyclicality of the endogenous component of government policy and the degree of Edgeworth complementarity between private consumption and government spending work in opposite directions in terms of generating a positive correlation between $\hat{y}$ and $\hat{g}$. Intuitively, Edgeworth complementarity between private consumption and government spending (i.e., $\alpha_g < 0$) tends to increase the correlation between $\hat{y}$ and $\hat{g}$, since under such a configuration an increase in government expenditures would induce people to consume more. At the same time, a countercyclical component in the policy rule reduces this correlation.

This yields a trade-off. Given an observed correlation between output and government spending, a highly countercyclical component of policy must be compensated by a high degree of Edgeworth complementarity. Conversely, if policy is exogenous, a lower degree of Edgeworth complementarity will suffice to match the observed pattern of correlation between output and government expenditures. This is illustrated in Figure 1.

Figure 1 shows two iso-correlation loci in the (φ_g, α_g) plane, depending on the relative sizes of the structural disturbances. Each point in these loci gives a particular (φ_g, α_g) combination resulting in the same correlation between output and government spending. When $\sigma_z > \sigma_u$, the iso-correlation locus is decreasing with φ_g , with a steep slope. This means that, as the degree of countercyclicality in the endogenous component of policy increases, it takes more and more complementarity to match the observed correlation. When $\sigma_u > \sigma_z$, the iso-correlation curve is much flatter, but the trade-off still exists.

This trade-off paves the way for a potential bias in the estimated degree of Edgeworth complementarity (and, by virtue of proposition 1, in the estimated multiplier). Suppose that an econometrician seeks to estimate α but uses a

FIGURE 1. ISO-CORRELATION LOCI IN THE (φ_g, α_g) PLANE

Notes: The iso-correlation loci are computed assuming, for illustrative purposes, $\nu = 5$, $s_g = 0.2$, and $\text{corr}(\hat{y}, \hat{g}) = 0.3$. In the $\sigma_u < \sigma_z$ case, we impose $\sigma_u/\sigma_z = 0.5$. In the $\sigma_u > \sigma_z$ case, we impose $\sigma_u/\sigma_z = 2$.

misspecified model in which φ_g is set to zero, while actually $\varphi_g > 0$. The above reasoning suggests that this would result in a downward-biased estimate of α , immediately translating into a downward-biased estimated multiplier. The next section formally establishes this.

C. The Effect of Omitting Endogenous Policy

Direct calculations yield the model's reduced form:

$$(7) \quad \hat{y}_t = \frac{\alpha\varphi_g}{1 + \alpha\varphi_g} \hat{y}_{t-1} + \frac{\alpha}{1 + \alpha\varphi_g} u_t + \frac{1}{1 + \alpha\varphi_g} z_t$$

$$(8) \quad \hat{g}_t = \frac{\varphi_g}{1 + \alpha\varphi_g} \hat{y}_{t-1} + \frac{1}{1 + \alpha\varphi_g} u_t - \frac{\varphi_g}{1 + \alpha\varphi_g} z_t.$$

From this reduced form, the structural parameters $(\alpha, \varphi_g, \sigma_u, \sigma_z)$ can be estimated consistently using the *plim* of the maximum likelihood estimation or an instrumental variable technique (with a relevant choice of instrumental variables).⁶

⁶ See the online Appendix.

Imagine the econometrician ignores the feedback rule and seeks to estimate the parameter α from data $(\hat{y}_t, \hat{g}_t)$ generated by model (7)–(8). The model considered by this econometrician is thus of the form

$$(9) \quad \hat{y}_t = \tilde{\alpha} \tilde{u}_t + \tilde{z}_t$$

$$(10) \quad \hat{g}_t = \tilde{u}_t.$$

By ignoring the parameter φ_g , the econometrician is implicitly estimating the government spending effects on output through a single-equation approach in a simultaneous-equation setup. As is well known from standard econometrics textbooks, she potentially faces a severe *simultaneous-equation* bias (see Greene 1997; Hamilton 1994).

To see this clearly, the ML estimator $\hat{\tilde{\alpha}}$ of $\tilde{\alpha}$ would be $E\{\hat{y}_t \hat{g}_t\}/E\{\hat{g}_t^2\}$, which simply corresponds to the OLS estimator. While φ_g exerts no influence on the long-run multiplier (see Proposition 1), the next proposition establishes that this parameter corrupts the estimated composite parameter $\tilde{\alpha}$ when policy is assumed to be exogenous.

PROPOSITION 2: *Under the previous hypotheses:*

(i) *The plim of the Maximum Likelihood estimator $\hat{\tilde{\alpha}}$ of $\tilde{\alpha}$ is*

$$(11) \quad \hat{\tilde{\alpha}} = \frac{\alpha(1 + \alpha\varphi_g)\sigma_u^2 - \varphi_g\sigma_z^2}{(1 + \alpha\varphi_g)\sigma_u^2 + 2\varphi_g^2\sigma_z^2}.$$

(ii) *Whenever $\sigma_z > 0$ and $\varphi_g > 0$, $\hat{\tilde{\alpha}}$ is downward-biased.*

(iii) *$\forall \sigma_z > 0$ and $\forall \sigma_u > 0$, we have*

(a) *If $\alpha \geq \sigma_z/\sigma_u$, then*

$$\frac{\partial}{\partial \varphi_g}(\hat{\tilde{\alpha}} - \alpha) < 0, \quad \forall \varphi_g \geq 0.$$

(b) *If $\alpha < \sigma_z/\sigma_u$, $\exists \bar{\varphi}_g(\alpha, \sigma_z, \sigma_u) > 0$, such that*

$$\frac{\partial}{\partial \varphi_g}(\hat{\tilde{\alpha}} - \alpha) \leq 0, \quad \forall \varphi_g \in [0, \bar{\varphi}_g(\alpha, \sigma_u, \sigma_z)],$$

$$\frac{\partial}{\partial \varphi_g}(\hat{\tilde{\alpha}} - \alpha) > 0, \quad \forall \varphi_g > \bar{\varphi}_g(\alpha, \sigma_u, \sigma_z).$$

PROOF:

See Appendix.

Part (i) of Proposition 2 establishes that the estimated value of $\hat{\tilde{\alpha}}$ is corrupted, in a nonlinear way, by φ_g , σ_u , and σ_z . The OLS regression does not pin down the effects of $\hat{g}$ on $\hat{y}$ but an average of private behavior and public policy, with weights that

depend on the relative size of the shocks' variances. Notice that this is more or less the formula displayed in standard econometrics textbooks in a demand-supply setup (e.g., see Hamilton 1994, chap. 9).

Part (ii) of Proposition 2 indicates under which circumstances omitting the endogenous component of government spending policy would result in a downward-biased estimate of α . Notice that under such circumstances, the long-run GSM is systematically downward-biased, by virtue of Proposition 1. Conversely, the only circumstances in which the bias vanishes are either $\varphi_g = 0$ or $\sigma_z = 0$. When $\varphi_g = 0$, the bias is obviously zero since, in this case, the model is well specified. If $\sigma_z = 0$, the bulk of fluctuations in $\hat{y}$ are accounted for by government spending shocks. In this case, the endogeneity bias vanishes. Endogenous public spending is positively related to the shock u and the (inverse) government policy shifts along the output equation.

When $\sigma_z > 0$ and $\sigma_u > 0$, the bias may increase or decrease with φ_g , depending on α , σ_z , and σ_u , as stated in part (iii) of Proposition 2. If $\alpha \geq \sigma_z/\sigma_u$, the bias increases. If $\alpha < \sigma_z/\sigma_u$, the bias increases with φ_g , up to a threshold value above which it decreases. However, the bias never reverts back to zero since $\lim_{\varphi_g \rightarrow \infty} \hat{\alpha} = 0 < \alpha$.

In the special case $\sigma_z > 0$ and $\sigma_u \rightarrow 0$, government spending shocks do not contribute much to the variance of $\hat{y}$. In this case, we obtain $\hat{\alpha} \rightarrow -1/(2\varphi_g)$. Thus, omitting the endogeneity of government spending would lead us to estimate a negative value of $\hat{\alpha}$. This is because endogenous public spending is negatively related to the shock z that shifts aggregate output. The covariance between $\hat{y}_t$ and $\hat{g}_t$ is negative and thus the estimated effect of public spending on output is negative. We are in the case when the output equation moves along the policy rule equation (which is truly downward sloping). In this case, the econometrician almost recovers the reverse government policy rule.

So far, we have focused on biases for the long-run GSM. However, omitting the endogeneity of the policy rule also has critical consequences on the estimated short-run responses of output to a government spending shock. In model (7)–(8), the effects of the government policy shock are long lasting provided $\varphi_g > 0$. Indeed, the dynamic response of output at $t + i$, for $i \geq 0$, after a shock on government spending at t is given by the formula

$$\frac{\alpha}{1 + \alpha\varphi_g} \left(\frac{\alpha\varphi_g}{1 + \alpha\varphi_g} \right)^i.$$

It is worth noticing that the cumulated dynamic responses are equal to α . Notice that as long as $\varphi_g > 0$, these impulse responses are always positive.

When the model is ill-specified as in (9)–(10), the estimated dynamic responses are $\hat{\alpha}$ on impact and zero after for $i > 0$. Consequently, the dynamic responses are underestimated for any period greater than zero. In addition, the cumulated responses are now given by $\hat{\alpha}$ and thus, following Proposition 2, the true cumulated dynamic responses are underestimated.

The response of $\hat{y}_t$ on impact is biased provided $\varphi_g > 0$. However, the sign of the bias is less trivial than for the long-run GSM. Indeed, φ_g affects the impact response of $\hat{y}_t$ in model (7)–(8). This response is given by $\alpha/(1 + \alpha\varphi_g)$ and thus it is a decreasing function of φ_g for $\alpha > 0$. At the same time, an econometrician

omitting the endogeneity of the government policy would obtain an impact response equal to $\hat{\alpha}$, which may be a decreasing function of φ_g too (see Proposition 2 and the related discussion). The next proposition characterizes the bias for the short-run response with respect to σ_u^2/σ_z^2 and α .

PROPOSITION 3: *Assume $\varphi_g > 0$. There exists a threshold value φ_g^* for φ_g such that the bias for the impact response of output to a government spending shock is zero. The estimated response is then downward-biased whenever $\varphi_g > \varphi_g^*$. This happens in the following cases:*

- (i) *If $\sigma_z < \sigma_u/\sqrt{3}$ and*
 - (a) $\alpha \in [0, (\sigma_z/\sigma_u)]$ *and $\varphi_g > 0 > \varphi_g^*$;*
 - (b) $\alpha \in [(\sigma_z/\sigma_u), \sqrt{3}(\sigma_z/\sigma_u))$ *and $\varphi_g > \varphi_g^* > 0$;*
 - (c) $\alpha \in (\sqrt{3}(\sigma_z/\sigma_u), 1]$ *and $\varphi_g > 0 > \varphi_g^*$.*
- (ii) *If $\sigma_z \in [\sigma_u/\sqrt{3}, \sigma_u]$ and*
 - (a) $\alpha \in [0, (\sigma_z/\sigma_u)]$ *and $\varphi_g > 0 > \varphi_g^*$;*
 - (b) $\alpha \in [(\sigma_z/\sigma_u), 1]$ *and $\varphi_g > \varphi_g^* > 0$.*
- (iii) *If $\sigma_z > \sigma_u$ and $\varphi_g > 0 > \varphi_g^*$.*

PROOF:

See Appendix.

The sign of φ_g^* depends on the value of α and on the relative variance of the two shocks σ_u^2/σ_z^2 . For example, when $\sigma_z < \sigma_u$, $\sigma_z > \sigma_u/\sqrt{3}$ and $\alpha \in [\sigma_z/\sigma_u, 1]$, φ_g^* is positive and the endogenous component of government policy must be sufficiently countercyclical to get a downward bias. This configuration corresponds to the situation where the government policy shock is the main driver of $\hat{y}_t$. Conversely, when $\sigma_z > \sigma_u$, φ_g^* is negative and the downward bias under the misspecified policy rule holds for a wider range of values for φ_g . This will happen when the shock to government policy accounts for a small portion of the variance of $\hat{y}_t$. Notice that a countercyclical component of policy is a sufficient condition to underestimate the true impact response of output.

II. A Quantitative Model

We now work out a quantitative extension of the previous model that we formally take to the data. We extend the previous setup by allowing for capital accumulation, habit formation in leisure decisions, and multiple shocks. While the model is arguably very stylized, it turns out to deliver a good fit.⁷

⁷ In the online Appendix, we modify the benchmark specification allowing for habits in consumption and dynamic adjustment costs and news shocks in the policy rule. Our results are robust to these perturbations.

A. The Model

The representative household's intertemporal expected utility is

$$(12) \quad E_t \sum_{i=0}^{\infty} \beta^i \left\{ e^{\epsilon_{c,t+i}} \log(c_{t+i} + \alpha_g g_{t+i}) - e^{\epsilon_{n,t+i}} \frac{\eta}{1+\nu} \left(\frac{n_{t+i}}{n_{t+i-1}^\phi} \right)^{1+\nu} \right\},$$

where $E_t\{\cdot\}$ denotes the expectation operator conditional on the information set at period t , and $\beta \in (0, 1)$ is the subjective discount factor. As in the previous section, the parameter α_g governs the substitutability/complementarity between private consumption and public expenditures.

The parameter ϕ governs the habit persistence in labor supply, and $\eta \geq 0$ is a scale parameter. When the parameter $\phi \neq 0$, labor supply decisions are subject to time nonseparabilities. If $\phi < 0$, labor supply displays intertemporal substitutability, whereas $\phi > 0$ implies intertemporal complementarity. Eichenbaum, Hansen, and Singleton (1988) showed that a specification with intertemporal complementarities is favored by the data. More recently, this specification has proven to be empirically relevant, as it translates habit persistence in leisure choices into aggregate output persistence (see Bouakez and Kano 2006; Dupaigne, Fève, and Matheron 2007; and Wen 1998). While other specifications that allow to capture the persistence in hours have been considered in the literature (e.g., adjustment costs on labor input, as in Chang, Doh, and Schorfheide 2007, or learning-by-doing, as in Chang, Gomes, and Schorfheide 2002), it turns out that the implied reduced form are almost identical to that resulting from our specification.

Utility derived from consumption is altered by a preference shock $\epsilon_{c,t}$, which obeys

$$\epsilon_{c,t} = \rho_c \epsilon_{c,t-1} + \sigma_c \zeta_{c,t},$$

where $|\rho_c| < 1$, $\sigma_c > 0$ and $\zeta_{c,t}$ is *independently and identically distributed* with $\zeta_{c,t} \sim N(0, 1)$. Labor disutility is subject to a preference shock $\epsilon_{n,t}$, which obeys

$$\epsilon_{n,t} = \rho_n \epsilon_{n,t-1} + \sigma_n \zeta_{n,t},$$

where $|\rho_n| < 1$, $\sigma_n > 0$ and $\zeta_{n,t}$ is *independently and identically distributed* with $\zeta_{n,t} \sim N(0, 1)$. As noted by Galí (2005), this shock accounts for a sizable portion of aggregate fluctuations. Moreover, it allows us to capture various distortions on the labor market, labeled *labor wedge* in Chari, Kehoe, and McGrattan (2007).

The representative household supplies hours n_t and capital k_t to firms, and pays a lump-sum tax T_t to the government. Accordingly, the representative household's budget constraint in every period t is

$$(13) \quad c_t + x_t + E_t\{q_{t,t+1} b_{t+1}\} = w_t n_t + r_t k_t - T_t + b_t,$$

where w_t is the real wage, r_t is the rental rate of capital, and x_t denotes investment.

The capital stock evolves according to

$$(14) \quad k_{t+1} = (1 - \delta)k_t + x_t,$$

where $\delta \in [0, 1]$ is the constant depreciation rate. As before, b_{t+1} represents the value in period $t + 1$ of the household's bond portfolio at the end of period t . As we assume complete financial markets, this portfolio includes state-contingent claims. The market value of such a bundle of state-contingent claims in period t is given by $E_t\{q_{t,t+1} b_{t+1}\}$, where the random variable $q_{t,t+1}$ is the stochastic discount factor for pricing arbitrary financial claims. The representative household thus maximizes (12) subject to the sequence of constraints (13) and (14), $t \geq 0$.

The representative firm produces a homogeneous final good y_t through the constant returns-to-scale technology

$$y_t = k_t^\theta (e^{z_t} n_t)^{1-\theta},$$

where k_t and n_t denote the inputs of capital and labor, respectively, $\theta \in (0, 1)$ is the elasticity of output with respect to capital, and z_t is a shock to total factor productivity, which follows a random walk process with drift of the form

$$z_t = \log(\gamma_z) + z_{t-1} + \sigma_z \zeta_{z,t},$$

where $\sigma_z > 0$ and $\zeta_{z,t}$ is independently and identically distributed with $\zeta_{z,t} \sim N(0, 1)$. The constant term $\gamma_z > 1$ is the drift term and accounts for the deterministic component of the growth process. Profit maximization implies $r_t = \theta y_t / k_t$ and $w_t = (1 - \theta) y_t / n_t$.

Government expenditures are entirely financed by taxes $T_t = g_t$. Notice that Ricardian equivalence holds in our setup, so that introducing government debt is unnecessary. The stationary component of government spending is given by

$$g_t e^{-z_t} = \bar{g}^s \tilde{g}_t e^{\epsilon_{g,t}},$$

where $\bar{g}^s$ denotes the deterministic steady-state value of $g_t e^{-z_t}$. The endogenous policy component $\tilde{g}_t$ obeys

$$\log(\tilde{g}_t) = -\varphi_g (\Delta \log(y_t) - \log(\gamma_z)),$$

and the stochastic (discretionary) component is assumed to follow an autoregressive process of the form

$$\epsilon_{g,t} = \rho_g \epsilon_{g,t-1} + \sigma_g \zeta_{g,t},$$

where $|\rho_g| < 1$, $\sigma_g > 0$, and $\zeta_{g,t}$ is *independently and identically distributed* with $\zeta_{g,t} \sim N(0, 1)$. Here, Δ stands for the first-difference operator. The parameter φ_g is the policy rule parameter linking the stationary component of government policy to demeaned output growth. If $\varphi_g < 0$, the policy rule contains a procyclical component that triggers an increase in government expenditures whenever output

growth is above its average value. In contrast, if $\varphi_g > 0$, the policy rule features a countercyclical component. In both cases, however, assessing the degree of pro cyclical or countercyclical of the overall level of government spending requires taking the stochastic trend in productivity into account. For example, assuming that $\varphi_g = 0$, the growth rate of government expenditures would still be positively correlated with total factor productivity growth.⁸

The homogeneous good can be used for private consumption c_t , government consumption g_t , and investment x_t . The market clearing condition on the goods market accordingly writes

$$y_t = c_t + x_t + g_t.$$

In the context of this model featuring a stochastic trend in productivity, we must modify Definition 1. To this end, we start by defining the detrended variables $y_t^s \equiv y_t e^{-z_t}$, $g_t^s \equiv g_t e^{-z_t}$, $c_t^s \equiv c_t e^{-z_t}$, and $x_t^s \equiv x_t e^{-z_t}$.

DEFINITION 2: *The long-run government spending multiplier, denoted by $\Delta y / \Delta g$, is the increase in steady-state, detrended output $\bar{y}^s$ after an increase in steady-state, detrended government spending expenditures $\bar{g}^s$*

$$\frac{\Delta y}{\Delta g} \equiv \frac{d\bar{y}^s}{d\bar{g}^s}.$$

Similarly, the long-run multiplier on consumption, denoted by $\Delta c / \Delta g$, and the long-run multiplier on investment, denoted by $\Delta x / \Delta g$, are defined by

$$\frac{\Delta c}{\Delta g} \equiv \frac{d\bar{c}^s}{d\bar{g}^s},$$

and

$$\frac{\Delta x}{\Delta g} \equiv \frac{d\bar{x}^s}{d\bar{g}^s},$$

respectively, where $\bar{c}^s$ and $\bar{x}^s$ are the steady-state values of detrended consumption and investment, respectively.

Using this definition, we can characterize how α_g and the long-run GSM are linked together. This relation is stated in the following proposition, which generalizes Proposition 1 to a setup with investment.

⁸ In Section III, we explore the robustness of our results to alternative specifications of the feedback rule.

PROPOSITION 4: *Under the preceding assumptions:*

(i) *The long-run government spending multiplier $\Delta y/\Delta g$ is*

$$\frac{\Delta y}{\Delta g} = \frac{1 - \alpha_g}{1 - s_x + \mu[1 - s_x - s_g(1 - \alpha_g)]},$$

where

$$\mu \equiv (1 + \nu)(1 + \phi) - 1, \quad s_x \equiv \frac{(\gamma_z - 1 + \delta)\theta\beta}{\gamma_z - \beta(1 - \delta)}.$$

(ii) *The long-run government multipliers $\Delta y/\Delta g$, $\Delta c/\Delta g$, and $\Delta x/\Delta g$ are linked through the relations*

$$\frac{\Delta c}{\Delta g} = (1 - s_x) \frac{\Delta y}{\Delta g} - 1$$

and

$$\frac{\Delta x}{\Delta g} = s_x \frac{\Delta y}{\Delta g}.$$

(iii) *The multiplier $\Delta y/\Delta g$ decreases with α_g .*

PROOF:

See Appendix.

Notice that when α_g and ϕ are set to zero, the above formula collapses to those reported in Baxter and King (1993) and Aiyagari, Christiano, and Eichenbaum (1992). As in the previous section, φ_g clearly does not show up in this formula. It remains to be seen whether omitting φ_g at the estimation stage can compromise the inference on α_g .

B. Data and Estimation

Before, taking the model to the data, we first induce stationarity by getting rid of the stochastic trend component z_t and we log-linearize the resulting system in the neighborhood of the deterministic steady state.⁹ Then, let $\hat{s}_t$ denote the vector collecting the log-linear model variables. The log-linear solution is of the form

$$(15) \quad \hat{s}_t = \mathbf{F}(\psi) \hat{s}_{t-1} + \mathbf{G}(\psi) \begin{pmatrix} \zeta_{z,t} \\ \zeta_{c,t} \\ \zeta_{n,t} \\ \zeta_{g,t} \end{pmatrix},$$

⁹ See Appendix for further details on the procedure used to induce stationarity.

where ψ is the vector of model's parameters. The system matrices $\mathbf{F}(\psi)$ and $\mathbf{G}(\psi)$ are complicated functions of the model's parameters.

We use as observable variables in estimation the logs of output, consumption, hours worked, and government expenditures. The measurement equation is

$$(16) \quad \begin{pmatrix} \Delta \log(y_t) \\ \log(n_t) \\ \Delta \log(c_t) \\ \Delta \log(g_t) \end{pmatrix} = \begin{pmatrix} \gamma_z - 1 \\ m_n(\psi) \\ \gamma_z - 1 \\ \gamma_z - 1 \end{pmatrix} + \mathbf{H} \hat{s}_t.$$

Here, $m_n(\psi)$ is a function that gives average log hours as a function of ψ and $\mathbf{H}$ is a selection matrix. For a given ψ , using equations (15) and (16), the log likelihood is evaluated via standard Kalman filter techniques. The estimated parameters are then obtained by maximizing the log likelihood.¹⁰

The quarterly data used for estimation come from the Federal Reserve Bank of St. Louis' FRED II database and from the Bureau of Labor Statistics website. They consist of government consumption expenditures and gross investment (GCE), private investment and private consumption, all deflated by the implicit GDP deflator (GDPDEF). Private investment is defined as the sum of gross private domestic investment (GPDI) and personal consumption expenditures on durable goods (PCDG). Private consumption is measured as the sum of personal consumption expenditures on non-durable goods (PCND) and services (PCESV). Output is then defined as the sum of private investment, private consumption, and government expenditures. Hours are borrowed from Francis and Ramey (2009). These hours data refer to the total economy and are adjusted for low-frequency movements due to changes in demographics, thus displaying less low-frequency behavior than unadjusted data. All the series are converted to per capita terms by dividing them by the civilian population, age 16 and over (CNP16OV). All the series are seasonally adjusted except for population. Our sample runs from 1960:I to 2007:IV.¹¹

The vector of parameters ψ is split in two subvectors ψ_1 and ψ_2 . The first one, $\psi_1 = (\beta, \delta, \nu, \theta, s_g)$, contains parameters calibrated prior to estimation. Typically, these are parameters difficult to estimate in our framework. The subjective discount factor, β , is set to 0.9951, yielding a real annual interest rate of 3.75 percent. The depreciation rate, δ , is set to 0.0153, to match the average investment-output ratio.

The parameter ν is set to 4 so that the inverse of the steady-state Frisch elasticity of labor supply $\mu \equiv (1 + \nu)(1 - \phi) - 1$ is close to 2 in the benchmark model, in accordance with previous studies (see Smets and Wouters 2007). Finally we set $\theta = 0.30$, so that the labor income share in output is 70 percent, and $s_g = 0.2$, so as to reproduce the average ratio of government expenditures to output in our sample.

¹⁰ We used different measurement equations, using the logged private consumption-output and logged government expenditures-output ratios instead of consumption growth and government expenditures growth. Estimation results were almost identical.

¹¹ In Section III, we consider the robustness of our results to alternative estimation samples.

The remaining parameters, contained in $\psi_2 = (\phi, \alpha_g, \gamma_z, \varphi_g, \rho_g, \rho_c, \rho_n, \sigma_z, \sigma_g, \sigma_c, \sigma_n)$, are estimated.¹² Estimation results are reported in Table 1. We consider four model restrictions, according to whether α_g and φ_g are constrained. Table 1 reports the log-likelihood $\mathcal{L}$ for each model specification, which we use naturally as our selection criterion. The restrictions are summarized below:

- Model (1): $\alpha_g = 0, \varphi_g = 0$, so that g has no direct effect on the marginal utility of private consumption and is exogenous.
- Model (2): $\alpha_g = 0, \varphi_g \neq 0$, so that g has no direct effect on the marginal utility of private consumption and is endogenous.
- Model (3): $\alpha_g \neq 0, \varphi_g = 0$, so that g has a direct effect on the marginal utility of private consumption and is exogenous.
- Model (4): $\alpha_g \neq 0, \varphi_g \neq 0$, so that g has a direct effect on the marginal utility of private consumption and is endogenous.

Overall, the model specifications yield precisely estimated parameters.¹³ Notice that, except for ρ_g and σ_g , most parameters are pretty invariant to the restrictions imposed on α_g or φ_g .

The log-likelihood comparison suggests the following comments. First, comparing specifications (1) with (3) or (2) with (4), one can clearly see that the restrictions $\alpha_g = 0$ is strongly rejected by the data. The p -values of the associated likelihood ratio tests are almost zero. In specifications (3) and (4), the parameter α_g is negative and significantly different from zero, consistent with the above likelihood ratio tests. This result is not an artifact of allowing for an endogenous government policy. This suggests that private and public consumption are complements. This result echoes findings by Karras (1994). Bouakez and Rebei (2007) and Mazraani (2010) also reach similar conclusions with a specification of preferences in which private consumption and government expenditures interact through a CES subutility function. We also used such a model version. As discussed above, since both representations yield the same log linear first-order conditions, this model version yields the exact same estimation results. However, the CES specification raises an identification problem that our specification eschews. It turns out that the government spending weight in utility is not separately identified in the CES case.

Second, the restriction $\varphi_g = 0$ is clearly rejected by the data. To see this, compare specifications (1) and (2) or (3) and (4). In both cases, the associated p -values are almost zero. These results strongly support the view that government policy comprises an endogenous component. To sum up, specification (4) is our preferred model. This specification thus features a positive effect of government spending on

¹² The parameter vector ψ_2 also contains $\bar{n}$, which is the average level of log hours. In our setup, estimating $\bar{n}$ is equivalent to estimating η . This parameter does not play any role in the log-linearized dynamics and is not reported.

¹³ We numerically checked the estimation convergence by shocking initial conditions on parameter values in the likelihood maximization step. Upon convergence, we also plotted slices of the likelihood function around each estimated parameter value to check local identification.

TABLE 1—ESTIMATION RESULTS

	Specification (1) $\alpha_g = 0, \varphi_g = 0$	Specification (2) $\alpha_g = 0, \varphi_g \neq 0$	Specification (3) $\alpha_g \neq 0, \varphi_g = 0$	Specification (4) $\alpha_g \neq 0, \varphi_g \neq 0$
α_g	—	—	−0.6340 (0.1357)	−0.9452 (0.1410)
φ_g	—	0.5723 (0.0803)	—	0.6117 (0.0660)
ϕ	0.4004 (0.0764)	0.4138 (0.0730)	0.3766 (0.0714)	0.4110 (0.0638)
γ_z	1.0043 (0.0007)	1.0043 (0.0007)	1.0043 (0.0007)	1.0044 (0.0007)
ρ_g	0.9469 (0.0087)	0.9592 (0.0078)	0.9535 (0.0079)	0.9756 (0.0055)
ρ_c	0.9762 (0.0049)	0.9775 (0.0047)	0.9795 (0.0036)	0.9834 (0.0031)
ρ_n	0.8346 (0.0424)	0.8345 (0.0411)	0.8366 (0.0366)	0.8399 (0.0330)
σ_z	0.0107 (0.0005)	0.0107 (0.0005)	0.0107 (0.0005)	0.0107 (0.0005)
σ_g	0.0130 (0.0007)	0.0120 (0.0006)	0.0129 (0.0007)	0.0119 (0.0006)
σ_c	0.0103 (0.0009)	0.0102 (0.0009)	0.0123 (0.0011)	0.0133 (0.0013)
σ_n	0.0270 (0.0014)	0.0266 (0.0014)	0.0273 (0.0015)	0.0266 (0.0014)
$\mathcal{L}$	2,655.3227	2,679.4375	2,665.9338	2,701.3173

Notes: Sample period: 1960:I–2007:IV. Standard errors in parentheses.

the marginal utility of private consumption and a countercyclical feedback effect in government spending.¹⁴

Notice that the estimated value for φ_g is large, suggesting a strong countercyclical component to policy. However, this should not necessarily reflect an extremely countercyclical policy. To see this, recall that φ_g governs the cyclicity of government expenditures in deviation from the stochastic TFP trend. To the extent that innovations to TFP account for a significant portion of fluctuations, the overall level of government expenditures need not be overly countercyclical, even when φ_g is large.

Even though all second order moments are given the same weight in the likelihood, comparing unconditional moments from the models to their empirical counterparts is useful to get some intuition on why model (4) is preferred. Results are reported in Table 2. We consider moments documenting the volatility, persistence, and comovement of key variables.

All the model versions perform equally well in terms of fitting standard errors of key aggregate variables. More interestingly, we see that whether or not $\alpha_g = 0$ is imposed, the correlation between changes in private consumption and changes in government spending is smaller when $\varphi_g > 0$. To see this, compare the results for specifications (1) against (2) or (3) against (4). This illustrates that in a standard

¹⁴ In the online Appendix, we use data simulated from specification (3) and (4) to establish that the negative empirical link found between α_g and φ_g is not idiosyncratic to our sample.

TABLE 2—MOMENTS COMPARISON

	Data	Specification (1) $\alpha_g = 0, \varphi_g = 0$	Specification (2) $\alpha_g = 0, \varphi_g \neq 0$	Specification (3) $\alpha_g \neq 0, \varphi_g = 0$	Specification (4) $\alpha_g \neq 0, \varphi_g \neq 0$
$\sigma(\Delta y)$	0.0093	0.0093	0.0093	0.0095	0.0095
$\sigma(\Delta c)$	0.0050	0.0071	0.0071	0.0074	0.0065
$\sigma(\Delta x)$	0.0327	0.0255	0.0281	0.0255	0.0308
$\sigma(\Delta g)$	0.0110	0.0170	0.0145	0.0168	0.0140
$\sigma(\Delta n)$	0.0066	0.0069	0.0068	0.0070	0.0070
$\rho(\Delta y)$	0.3200	0.1165	0.1226	0.1116	0.1304
$\rho(\Delta c)$	0.2482	0.0217	0.0232	0.0176	0.1980
$\rho(\Delta x)$	0.2057	0.1599	0.0413	0.1425	−0.0681
$\rho(\Delta g)$	0.0948	−0.0161	0.1076	−0.0139	0.1337
$\rho(\Delta n)$	0.3886	0.3299	0.3437	0.3100	0.3511
$\text{corr}(\Delta y, \Delta c)$	0.5115	0.6336	0.6461	0.7282	0.6541
$\text{corr}(\Delta y, \Delta x)$	0.9043	0.8025	0.8567	0.7315	0.8246
$\text{corr}(\Delta y, \Delta g)$	0.2913	0.5616	0.3354	0.5873	0.3697
$\text{corr}(\Delta c, \Delta g)$	0.2368	0.1999	0.0204	0.4976	0.4001
$\text{corr}(\Delta x, \Delta g)$	−0.0288	0.1719	0.0209	0.0213	−0.1001

Notes: Sample period: 1960:I–2007:IV. $\sigma(\cdot)$, $\rho(\cdot)$, and $\text{corr}(\cdot, \cdot)$ stand for standard deviation, first-order autocorrelation coefficient, and correlation, respectively. Δ is the first difference operator, y is output, c is consumption, x is investment, g is government expenditures, n is hours worked.

neoclassical growth model, allowing for a countercyclical component of government spending policy works toward reducing the correlation between consumption growth and government spending shocks. Similarly, comparing the same specifications, we see that relaxing the constraint $\varphi_g = 0$ decreases the correlation between output growth and government expenditures growth. Conversely, whether or not $\varphi_g = 0$, relaxing the constraint $\alpha_g = 0$ increases the correlation between consumption growth and government spending growth. To see this, compare the results for specifications (1) against (3) or (2) against (4). This once again illustrates how Edgeworth complementarity and a countercyclical component of policy interact in our model. However, the correlation between private consumption growth and government spending growth is too strong in specification (4) compared to the data. At the same time, only this specification reproduces the negative correlation between investment growth and government spending growth. This negative correlation reflects the effects of Edgeworth complementarity; when government expenditures increase, people are willing to consume more, at least in the short run. This translates into reduced savings, thus yielding a relative decline in investment. Finally, we also see that, irrespective of constraints imposed on α_g , the restriction $\varphi_g = 0$ deteriorates the model’s ability to capture the persistence of changes in government expenditures and in private consumption.

Having explored the empirical properties of model (4), we now use this version to investigate the quantitative effects of omitting the government feedback rule.¹⁵

¹⁵ In the online Appendix, we also perform several specification tests on our models. The bottom line is that our benchmark model shows no clear sign of misspecification.

C. Quantitative Implications for Government Spending Multipliers

In this section, we assess the quantitative impact of omitting the countercyclical component of government expenditures policy. We proceed in two steps. First, guided by the analytical results obtained in the previous section, we investigate the consequences for the long-run GSM of misspecifying the policy rule. Second, we illustrate that these effects are also present in an assessment of shorter-term multipliers.

Long-Run Multiplier.—Upon inspecting models (3) and (4), we see that imposing $\varphi_g = 0$ strongly affects the estimated value of α_g . When φ_g is freely estimated, we obtain $\alpha_g = -0.95$, whereas we get $\alpha_g = -0.63$ when we impose $\varphi_g = 0$. Importantly, these parameter estimates are significantly different from each other, according to a standard Wald test.

What does this imply for the long-run government spending multiplier? To answer this question, we use the formula in Proposition 3 to estimate the long-run GSM. The estimated multipliers are reported in Table 3, together with their standard errors. When α_g is restricted to zero, the estimated long-run multiplier is typically less than one, as obtained in standard models (see Baxter and King 1993; Aiyagari, Christiano, and Eichenbaum 1992). Concretely, depending on the restrictions imposed on φ_g , we obtain values roughly comprised between 0.53 and 0.55. Importantly, comparing specifications (1) and (2) in Table 3, one can clearly see that φ_g has almost no discernible effect on the long-run multiplier. The reason why is simple: the parameters β , δ , and ν are restricted prior to estimation. In addition, the estimation results show that ϕ is relatively insensitive to the different specifications (see Table 1).

In our framework, omitting the feedback effect in the policy rule can impact on the long-run output multiplier only when the parameter α_g is freely estimated. This is the novel feature of our model, because the link between the policy feedback parameter (φ_g) and the degree of Edgeworth complementarity between public and private consumptions (α_g) could obviously not be studied in frameworks imposing $\alpha_g = 0$. The columns associated with specifications (3) and (4) in Table 3 thus give new results. More precisely, in model (3), the output multiplier is 0.97, while it reaches 1.31 in model (4). These values are significantly different from each other at conventional levels. As explained above, the higher multiplier in (4) derives from a smaller α_g than in (3).¹⁶

To complement on these results, we consider the following exercise. We set the feedback parameters φ_g to values on a grid between $\varphi_g = 0$ and the estimated value obtained in specification (4) (i.e., $\varphi_g = 0.6117$). For each value, all the remaining parameters in ψ_2 are reestimated. The results are reported on Figure 2.

The upper left panel reports the log likelihood as a function of φ_g . The gray area corresponds to restrictions on φ_g that are not rejected at the 5 percent level according to a likelihood ratio test. This gray area is also reported in each of the other panels

¹⁶ Similar results obtain for the multiplier on private investment. This is not surprising since this multiplier is strictly proportional to $\Delta y / \Delta g$ and the proportionality factor does not depend on α_g . The multiplier on private consumption is not significantly different from zero.

TABLE 3—ESTIMATED MULTIPLIERS

	Specification (1) $\alpha_g = 0, \varphi_g = 0$	Specification (2) $\alpha_g = 0, \varphi_g \neq 0$	Specification (3) $\alpha_g \neq 0, \varphi_g = 0$	Specification (4) $\alpha_g \neq 0, \varphi_g \neq 0$
$\Delta y / \Delta g$	0.5319 (0.0606)	0.5429 (0.0603)	0.9738 (0.1410)	1.3128 (0.1803)
$\Delta c / \Delta g$	-0.5955 (0.0461)	-0.5872 (0.0458)	-0.2596 (0.1072)	-0.0020 (0.1370)
$\Delta x / \Delta g$	0.1274 (0.0146)	0.1301 (0.0145)	0.2333 (0.0339)	0.3148 (0.0433)

Notes: Sample period: 1960:I–2007:IV. Standard errors in parentheses.

in Figure 2. The upper right panel reports the estimated value of α_g as a function of φ_g . The bottom panels report the long-run multipliers on output and consumption. Figure 2 makes clear that even loose restrictions on φ_g (i.e., restrictions not too far from the estimated value) are easily rejected and rapidly translate into higher α_g and much lower multipliers. Importantly, the continuous and decreasing mapping from φ_g to α_g (and thus on long-run multipliers) echoes the analytical findings obtained in the simple model explored in the first section.

To sum up, there exists a strong interaction between the estimated values of φ_g and α_g that have potentially dramatic implications for the quantitative assessment of the long-run government spending multiplier. We argued previously that this bias could also corrupt shorter term multipliers. The next section provides a quantitative illustration of this.

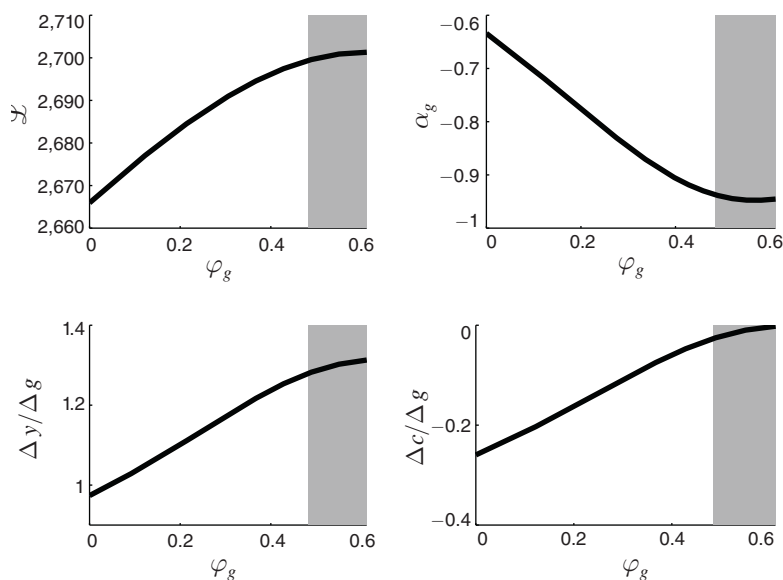
Short-Run Multipliers.—So far, we have insisted on the long-run GSM. The main reason for this is that this number does not directly depend on the serial correlation of government policy. Focusing on this multiplier thus illustrates in an unambiguous way how misspecifying the policy rule can result in a downward biased multiplier. However, all our results should in principle hold equally in an assessment of shorter-term multipliers.

To illustrate this, we follow Uhlig (2010) and specify an autoregressive process for the exogenous component of government spending designed to approximate the time profile of the ARRA plan. More precisely, we assume that the discretionary component to policy follows the process

$$\epsilon_{g,t} = 1.653 \epsilon_{g,t-1} - 0.672 \epsilon_{g,t-2},$$

which is initialized by setting $\epsilon_{g,0} = 0$ and $\epsilon_{g,1} = 0.32$ of detrended, steady-state output, assuming that date $t = 1$ corresponds to 2009:II. As argued by Uhlig (2010), this process closely approximates the government spending path reported in Cogan et al. (2010) (see, in particular, Uhlig 2010, Figure 1).

This process is fed into three model versions that we use to compute different versions of the short-run multiplier. First, we use specification (3), which imposes an exogenous policy at the estimation stage. Second, we use our preferred specification (4), in which policy parameters are freely estimated. Finally,

FIGURE 2. SENSITIVITY TO CONSTRAINTS ON POLICY RULE PARAMETER φ_g

Notes: Sample period: 1960:I–2007:IV. The feedback parameters φ_g take values on a grid between 0 and 0.6117. For each value, all the remaining parameters in ψ_2 are reestimated. The upper left panel reports the log likelihood as a function of φ_g . The gray area corresponds to restrictions on φ_g that are not rejected at the 5 percent level according to a likelihood ratio test. This gray area is also reported in each of the other panels. The upper right panel reports the estimated value of α_g as a function of φ_g . The bottom panels report the long-run multipliers on output and consumption.

we freeze all nonpolicy parameters obtained in this specification and impose an exogenous policy rule (i.e., $\varphi_g = 0$). We do this to make sure that allowing for a countercyclical policy rule does not change the resulting short-run multiplier too much. In this exercise, it is important to impose the same process for g_t in all our specifications. Indeed, as has been shown by Aiyagari, Christiano, and Eichenbaum (1992), the short-run multiplier heavily depends on the persistence of government expenditures. By freezing this process in all our experiments, we make sure that any difference in the government spending multiplier results from the estimated transmission mechanisms and/or the degree of cyclicity of the endogenous component of policy.

The results are reported in Figure 3. In the latter, the multiplier at $t + k$, $k \geq 0$ is defined as $\Delta y_{t+k} / \Delta g_t$, where the response of output is obtained under the assumed process for g_t . This is the definition of the short-run multiplier used in Galí, López-Salido, and Vallés (2007). The thick dark line corresponds to the multiplier under specification (4). The dashed dark line is the multiplier under an exogenous policy in specification (4), which we implement by setting $\varphi_g = 0$. Finally, the dotted line is the multiplier under specification (3). As a benchmark, we also report the multiplier that would obtain in the plain RBC model (i.e., specification (1)) which corresponds to the circles in Figure 3. As Figure 3 makes clear, with or without systematic policy, the multiplier is always much higher in specification (4) than in specification (3). Notice also that under

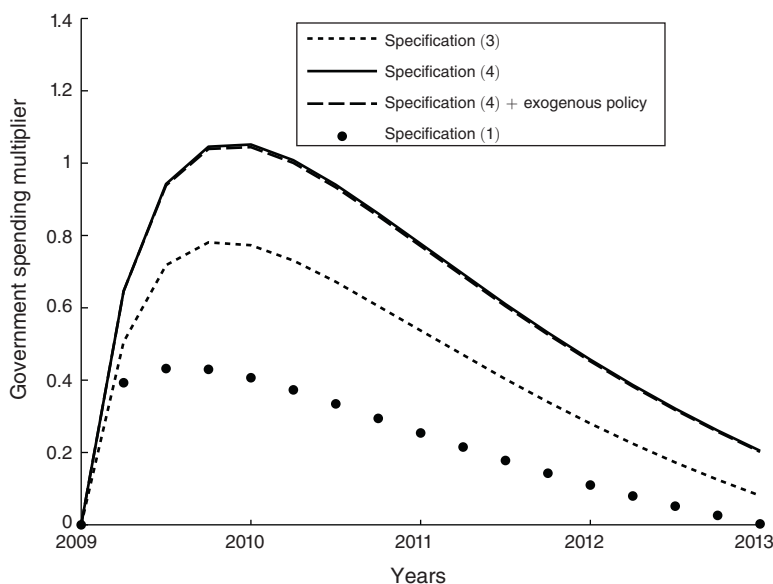


FIGURE 3. STIMULUS EFFECT ON OUTPUT

Notes: Estimated dynamic effects of government purchases in the February 2009 stimulus legislation on output, in relative deviation from steady state. Specification (3): $\alpha_g \neq 0$ and $\varphi_g = 0$; specification (4): $\alpha_g \neq 0$ and $\varphi_g \neq 0$; in specification (4) with $\varphi_g = 0$, we use the estimated value of α_g in (4) and impose the counterfactual $\varphi_g = 0$; specification (1): $\alpha_g = 0$ and $\varphi_g = 0$.

specification (1), which ignores the transmission mechanism of public expenditures as well as the countercyclical component of policy, the multiplier would be even lower.

As Figure 3 shows, the maximal multiplier is obtained approximately at the same horizon under the alternative specifications (about three quarters after the initial impulse). However, at the maximal effect, we obtain a difference of slightly more than 0.25 points between specifications (3) and (4). In other words, assuming an exogenous policy would yield a 25 cent bias in the short-run multiplier. This downward bias echoes analytical results on the impact response of output derived in our simple model.

Notice finally that the exact value of φ_g used in simulating specification (4) does not affect the multiplier quantitatively. This is consistent with our previous long-run findings.¹⁷ As argued before, this parameter plays a key role only at the estimation stage. Omitting it would lead an econometrician to downward bias the degree of Edgeworth complementarity, resulting in a seriously flawed assessment of the ARRA impact.

¹⁷ Notice, however, that if we were to impose $\alpha_g = \hat{\alpha}_g$ and $\varphi_g = 0$ in model (4), the fit would be dramatically deteriorated. The associated log likelihood would now be equal to 2663.3995. With such a restriction, the model would generate too much procyclical movements between public spending and both output and private consumption.

III. Robustness

In this section, we investigate the robustness of our results to alternative specifications for the government spending rule, alternative estimation subsamples, and an alternative model specification very close to Smets and Wouters (2007).

A. Alternative Specifications of the Feedback Rule

In this paper, the assumed fiscal rule lacks strong theoretical underpinnings, contrary to the celebrated Taylor rule for monetary policy. It is thus central, at the very least, to establish that our specification can be defended on empirical grounds. To this end, we reestimate the model under a large battery of alternative government spending rules. We then use the likelihood to compare these specifications.

The alternative rules considered in this section are specified as follows:

- (A) $\log(\tilde{g}_t) = -\varphi_g(\log(y_t) - z_t - \log(\bar{y}^s))$
- (B) $\log(\tilde{g}_t) = -\varphi_g(\log(y_{t-1}) - z_{t-1} - \log(\bar{y}^s))$
- (C) $\log(\tilde{g}_t) = -\varphi_z(\Delta z_t - \log(\gamma_z))$
- (D) $\log(\tilde{g}_t) = -\varphi_z(\Delta z_t - \log(\gamma_z)) - \varphi_c \Delta \epsilon_{c,t} - \varphi_n \Delta \epsilon_{n,t}$
- (E) $\log(\tilde{g}_t) = -\varphi_g(\Delta \log(y_t) - \log(\gamma_z)) - \varphi_z(\Delta z_t - \log(\gamma_z))$
- (F) $\log(\tilde{g}_t) = -\varphi_g(\Delta \log(y_t) - \log(\gamma_z)) - \varphi_z(\Delta z_t - \log(\gamma_z)) - \varphi_c \Delta \epsilon_{c,t} - \varphi_n \Delta \epsilon_{n,t}$
- (G) $\log(\tilde{g}_t) = -\varphi_g(\Delta \log(y_{t-1}) - \log(\gamma_z))$
- (H) $\log(\tilde{g}_t) = -\varphi_g(\lambda(\Delta \log(y_t) - \log(\gamma_z)) + (1 - \lambda)(\Delta \log(y_{t-1}) - \log(\gamma_z)))$,

where, as before, $\bar{y}^s$ denotes the steady-state value of detrended output.

Results are reported in Table 4. For comparison purposes, we reproduce the results obtained with our preferred specification (4), referred to here as the benchmark specification. The other specifications are: (A) the stationary component of government spending reacts to current deviations of output from its stochastic trend; (B) the stationary component of government spending reacts to once-lagged deviations of output from its stochastic trend; (C) the stationary component of government spending reacts to changes in total factor productivity, resembling the specification used by Smets and Wouters (2007); (D) the stationary component of government spending reacts to changes in all the structural shocks; (E) combines our benchmark specification with (C); (F) combines the benchmark specification with (D). We also consider an alternative rule (G) to our benchmark specification where we assess the effect of having $\log(\tilde{g}_t)$ respond to lagged output growth. With this specification, automatic stabilizers work with some delay. Finally, specification (H) nests our benchmark and (G), where the parameter λ puts a weight on contemporaneous output growth.¹⁸

¹⁸ We also considered versions of the fiscal rule with hours growth $\Delta \log(n_t)$ instead of output growth. We basically obtain the same results as in our benchmark specification.

TABLE 4—ALTERNATIVE GOVERNMENT SPENDING RULES

Specification	α_g	φ_g	λ	φ_z	φ_c	φ_n	$\Delta y/\Delta g$	$\mathcal{L}$
Benchmark	-0.9452	0.6117	—	—	—	—	1.3128	2,701.3173
(A)	-0.5774	-0.4866	—	—	—	—	0.9652	2,669.6339
(B)	-0.6287	0.1029	—	—	—	—	0.9623	2,666.1522
(C)	-0.7091	—	—	0.3952	—	—	1.0693	2,691.5604
(D)	-0.7893	—	—	0.4254	0.0133	-0.0500	1.1105	2,694.4357
(E)	-1.0398	0.8176	—	-0.1660	—	—	1.4244	2,702.0942
(F)	-1.2220	2.5102	—	-1.4928	-0.1132	0.2264	1.5455	2,714.2522
(G)	-0.6342	-0.0104	—	—	—	—	0.9736	2,665.9424
(H)	-1.0730	1.0802	0.7023	—	—	—	1.4514	2,712.7980
Exogenous policy	-0.6340	—	—	—	—	—	0.9738	2,665.9338

Notes: Sample period: 1960:I–2007:IV. Benchmark is specification (4): $\alpha_g \neq 0$ and $\varphi_g \neq 0$. Exogenous policy is specification (3): $\alpha_g \neq 0$ and $\varphi_g = 0$.

Table 4 reports the estimated values of α_g and the policy rule parameters, together with the implied long-run multipliers and the log likelihood. To ease comparison, we also report the estimation results obtained under an exogenous policy (i.e., specification (3)).

As is clear from Table 4, our benchmark specification dominates the alternative specifications from (A) to (D). Notice that these specifications are not nested with our benchmark. Since our benchmark implies a higher log likelihood, the alternative rules are not better descriptions of the data. Moreover, the specification test results deteriorate (not reported), especially so in terms of serial correlation of innovations.

The specifications that yield the lowest fit are (A) and (B). Specification (A) implies a procyclical component of government spending policy. In this case, the implied α_g is higher than under an exogenous policy (-0.58 in specification (A) and -0.63 under an exogenous policy). This mechanically translates into a smaller multiplier in (A), as expected from our previous analysis. In specification (B), the endogenous component of policy is almost acyclical and we obtain roughly similar multipliers than under an exogenous policy.¹⁹

Specification (C) implies a countercyclical component of policy in response to technology shocks, since the estimated values of φ_z is positive. Once again, as expected, the estimated α_g is slightly lower than under an exogenous policy ($\alpha_g = -0.71$), resulting in a higher multiplier.

Specification (D) adds to the former case by allowing policy to respond to all the shocks. This specification implies a countercyclical component of policy, in the sense that the estimated values of φ_z and φ_c are positive, while the estimated value of φ_n is negative. As expected, the Edgeworth complementarity parameter turns out to be lower than under an exogenous policy ($\alpha_g = -0.79$).

¹⁹ In the online Appendix, we use our benchmark model as a DGP and estimate on simulated data a misspecified model assuming that the government spending rule takes the form (A). While government policy is indeed countercyclical in the DGP, estimation outcomes under the misspecified model result, on average, in a procyclical policy rule. Interestingly, the modal estimate of φ_g in this experiment is very close to that obtained when estimating the model with rule (A) on actual data. We use this result to set up an encompassing test which leads us to reject specification (A) in favor of specification (4).

Specifications (E) and (F), which nest our benchmark, imply higher log likelihoods, by construction. A likelihood ratio test would not reject our benchmark when compared to specification (E). In contrast, the log-likelihood is much higher in specification (F). However, specification tests outcomes do not improve much when compared to our benchmark.

It is not obvious a priori to tell whether or not specifications (E) and (F) feature countercyclicality of the endogenous component of government spending policy. However, if our claim holds in this context, one can interpret the low value obtained for α_g as suggestive of a high degree of countercyclicality.

Estimation results show that specification (G) is clearly dominated by our benchmark representation. The log likelihood is close to that obtained under an exogenous policy. Specification (H) confirms this finding. When contemporaneous and once-lagged output growth appear both in the rule, our estimation results clearly favor a contemporaneous response of government spending (the weight λ is equal to 0.7023). Notice that in this case, the degree of Edgeworth complementarity is more pronounced and, accordingly, the model yields a higher multiplier.

B. *Subsample Analysis*

Perotti (2005) showed that empirical measures of government spending multipliers can prove sensitive to the particular sample selected. Importantly, he argues that multipliers over the sample 1980:I–2007:IV are smaller than those found over the sample 1960:I–1979:IV. We investigate whether our results still hold if we reestimate our model over the same subsamples.

Results are reported in Table 5. Our previous conclusions are broadly confirmed. First, the restriction $\varphi_g = 0$ is rejected, suggesting that government spending policy is endogenous, irrespective of the selected sample. Second, when this restriction is imposed, we obtain a higher α_g , resulting in a smaller multiplier. This holds over both subsamples. We also obtain a smaller long-run GSM over the second subsample, confirming results in Perotti (2005).

C. *A Smets-Wouters Type Model*

It has become common in macroeconomics to estimate richer models with many more shocks and frictions. As a final robustness check, we thus investigate whether our results hold within a medium-scale DSGE model along the lines of Christiano, Eichenbaum and Evans (2005) and Smets and Wouters (2007). The model combines a neoclassical growth core with several shocks and frictions. It includes features such as habit formation, investment adjustment costs, variable capital utilization, monopolistic competition in goods and labor markets, and nominal price and wage rigidities, which help to replicate the data. The economy is populated by five classes of agents: producers of a final good, intermediate goods producers, households, employment agencies, and a government. We adopt the specification investigated by Justiniano, Primiceri, and Tambalotti (2010), except that we allow for Edgeworth complementarity/substitutability and endogeneity of public spending. On top of these refinements, following Cogan et al. (2010), we consider two model versions depending on whether we allow

TABLE 5—SUBSAMPLE ROBUSTNESS

Specification	α_g	φ_g	ϕ	$\Delta y/\Delta g$	$\mathcal{L}$
1960:I–1979:IV					
(4)	−1.0167	0.6033	0.3755	1.3263	1,101.3585
(3)	−0.7910	—	0.3896	1.1276	1,088.1722
1980:I–2007:IV					
(4)	−0.8068	0.5597	0.4146	1.1856	1,614.9772
(3)	−0.6100	—	0.3777	0.9580	1,599.7944

Notes: Specification (4): $\alpha_g \neq 0$ and $\varphi_g \neq 0$. Specification (3): $\alpha_g \neq 0$ and $\varphi_g = 0$.

for non-Ricardian agents. Indeed, as shown by Galí, López-Salido, and Vallés (2007), allowing for liquidity-constrained agents has proven to be a useful modeling device for reproducing the aggregate effect of government spending shocks. In the remainder, we let ω denote the fraction of non-Ricardian agents.

We follow the Bayesian approach to estimate the log-linearized model. The dynamic system is cast in a state-space representation for the set of observable variables. The Kalman filter is then used to measure the likelihood of the observed variables and to form the posterior distribution of the structural parameters by combining the likelihood function with a joint density characterizing some prior beliefs. Given the specification of the model, the posterior distribution cannot be recovered analytically but may be computed numerically, using a Monte-Carlo Markov Chain (MCMC) sampling approach. More specifically, we rely on the Metropolis-Hastings algorithm to obtain a random draw of size 400,000 from the posterior distribution of the parameters. We estimate the model using seven series of US quarterly data: consumption growth, investment growth, real government expenditures growth, the growth rate of real hourly compensation, the logarithm of hours worked, the inflation rate, and the nominal interest rate. As before, real variables are divided by the civilian population, age 16 and over. Once again, all the data come from the Federal Reserve Bank of St. Louis’ FRED II database and from the Bureau of Labor Statistics website, except hours worked that are borrowed from Francis and Ramey (2009). Recall that the sample runs from 1960:I to 2007:IV.

We consider four model versions, labeled (3), (4), (3’), and (4’). As before, in (3), we impose $\varphi_g = 0$ while in (4) this parameter is freely estimated. In both specifications, we also impose that the mass of non-Ricardian agents be zero (i.e., $\omega = 0$). In specifications (3’) and (4’), we freely estimate ω , while considering $\varphi_g = 0$ in (3’) and $\varphi_g \neq 0$ in (4’). To save space, we only report estimation results for ω , φ_g , and α_g .²⁰ Importantly, we specify for φ_g and α_g uniform priors centered on zero and with a standard deviation of 0.65 and 1.30, respectively, to reflect our agnostic view concerning the sign of these key parameters.

Table 6 reports the posterior means and the associated 90 percent confidence interval for the parameters discussed above. Upon inspecting model (3) and (4), we see again that imposing $\varphi_g = 0$ strongly affects the estimated value of α_g : when φ_g is freely estimated, we obtain $\alpha_g = -0.63$, whereas we get $\alpha_g = -0.20$ when we

²⁰ Model details and full estimation results are relegated in the online Appendix.

TABLE 6—ESTIMATION RESULTS FOR THE SMETS-WOUTERS TYPE MODEL

Specification	ω	α_g	φ_g	$\Delta y/\Delta g$	Marginal likelihood
(4)	—	−0.626 [−0.914, −0.332]	0.752 [0.545, 0.964]	0.773 [0.546, 1.027]	−1,277.683
(3)	—	−0.199 [−0.454, 0.040]	—	0.489 [0.350, 0.639]	−1,298.946
(4')	0.149 [0.085, 0.211]	−1.133 [−1.443, −0.819]	0.728 [0.528, 0.919]	0.856 [0.639, 1.086]	−1,275.890
(3')	0.165 [0.095, 0.231]	−0.870 [−1.201, −0.552]	—	0.588 [0.431, 0.750]	−1,296.944

Notes: Specification (3): $\omega = 0$, $\alpha_g \neq 0$ and $\varphi_g = 0$; Specification (4): $\omega = 0$, $\alpha_g \neq 0$ and $\varphi_g \neq 0$; Specification (3'): $\omega \neq 0$, $\alpha_g \neq 0$ and $\varphi_g = 0$; Specification (4'): $\omega \neq 0$, $\alpha_g \neq 0$ and $\varphi_g \neq 0$. Figures in brackets are the cutoff values for the fifth and ninety-fifth percentile of the posterior distributions.

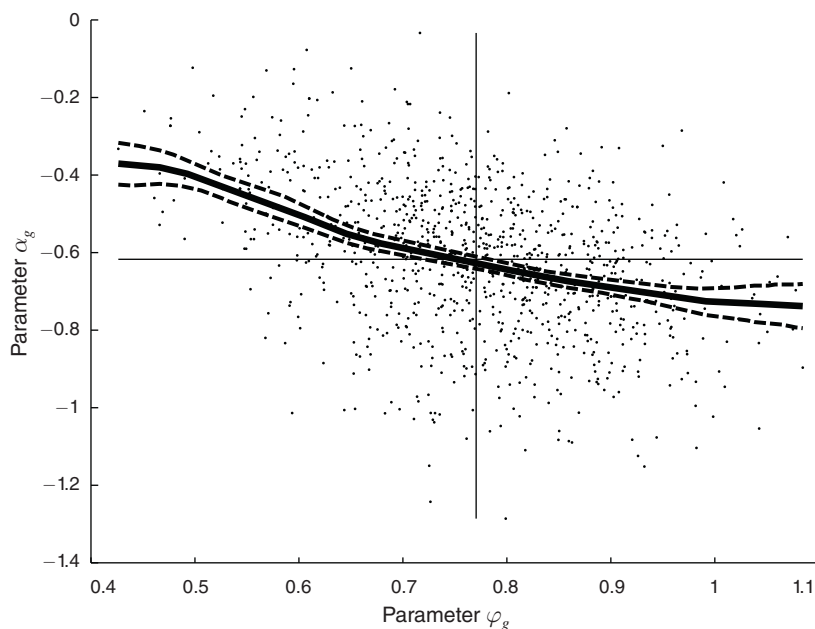
set $\varphi_g = 0$. Table 6 makes clear that ignoring the feedback rule would lower the degree of Edgeworth complementarity. As a consequence, this would also lead to a lower long-run GSM. It is worth noting that the implied difference in multipliers is very close to that reported in Table 3. All these findings confirm the results obtained from our benchmark model. In addition, the marginal likelihood strongly favors the model version with endogenous policy.

When considering specifications (3') and (4'), we obtain a small fraction of non-Ricardian agents. Notice that the presence of these agents does not change our basic message: the prior mean of α_g is smaller in absolute value when φ_g is constrained to zero. Once again, the marginal likelihood points to specification (4') as the preferred model version.

Finally, we plot draws from the posterior distribution of (i) φ_g and α_g , and (ii) φ_g and $\Delta y/\Delta g$ for specification (4). Figures 4 and 5 report the outcome of this exercise. The thick plain line is the nonparametric regression, and the thick dashed lines delineate the 90 percent confidence interval obtained by standard bootstrap techniques. The scatter diagram corresponds to the estimation of model (4). The cross indicates the average parameter values for φ_g and α_g in Figure 4, and the average parameter values for φ_g and $\Delta y/\Delta g$, in Figure 5. Figures 4 and 5 clearly reveal, in the neighborhood of the posterior means, the existence of (i) a negatively sloped relation between φ_g and α_g and (ii) a positively sloped relation between φ_g and $\Delta y/\Delta g$. Once again, these additional quantitative exercises corroborate findings from the simple analytical model and those from the benchmark setup.

IV. Conclusion

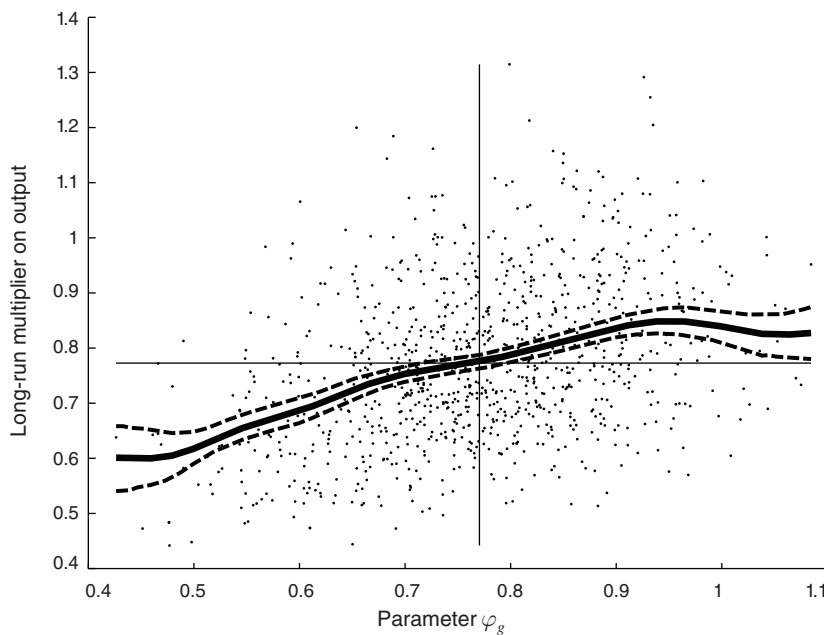
This paper has proposed to assess quantitatively the consequences of misspecifying the government spending rule on the estimated government spending multiplier within a DSGE framework. We first considered a simplified model version to show analytically that omitting the feedback rule at the estimation stage yields a downward-biased estimate of the long-run government spending multiplier. To establish this, we first showed that the multiplier is an increasing function of the

FIGURE 4. RELATIONSHIP BETWEEN φ_g AND α_g

Notes: The thick plain line is the nonparametric regression and the thick dashed lines delineate the 90 percent confidence interval obtained by standard bootstrap techniques. The cross indicates the average parameter values for α_g and φ_g .

degree of Edgeworth complementarity. In turn, complementarity and countercyclicality of the endogenous component of policy interact through cross-equation restrictions, paving the way for a potential bias. This result holds for other commonly considered transmission mechanisms (see the online Appendix). We then estimated on postwar US data a quantitative model version and obtained that omitting the endogeneity of government spending exerts a severe, downward impact on the estimated long-run multiplier. Such a bias also characterizes shorter-term multipliers and thus can seriously affect the quantitative assessment of fiscal packages such as the ARRA stimulus. Our results appear to be very robust to a series of perturbations to the benchmark specification.

In our framework, we have deliberately abstracted from relevant details in order to highlight, as transparently as possible, the empirical link between policy rule parameters and the degree of Edgeworth complementarity between private and public consumption. However, the recent literature insists on other modeling issues that might potentially affect our results. We mention two of them. First, as put forth by Leeper, Plante, and Traum (2010), a more general specification of government spending rule, lump-sum transfers, and distortionary taxation is needed to properly fit US data. This richer specification includes, in addition to the automatic stabilizer component, a response to government debt and comovement between tax rates. An important quantitative issue may be to assess which type of stabilization (automatic stabilization and/or debt stabilization) interacts with the estimated degree of Edgeworth complementarity. Second, Fiorito and Kollintzas (2004) have suggested

FIGURE 5. RELATIONSHIP BETWEEN φ_g AND $\Delta y/\Delta g$

Notes: The thick plain line is the nonparametric regression and the thick dashed lines delineate the 90 percent confidence interval obtained by standard bootstrap techniques. The cross indicates the average parameter values for α_g and φ_g .

that the degree of complementarity/substitutability between government and private consumptions is not homogeneous over types of public expenditures. This suggests to disaggregate government spending and inspect how feedback rules affect the estimated degree of Edgeworth complementarity in this more general setup. These issues will constitute the object of future researches.

APPENDIX

PROOF OF PROPOSITION 1:

To prove the first part of Proposition 1, evaluate equation (4) in the deterministic steady state, which implies

$$(A1) \quad \eta \bar{y}^\nu = \frac{1}{\bar{y} - (1 - \alpha_g)\bar{g}},$$

where $\bar{y}$ and $\bar{g}$ are the steady-state values of y and g , respectively.

Total differentiation of the above equation then yields

$$(A2) \quad \nu d\bar{y} = -\frac{\bar{y}}{\bar{y} - (1 - \alpha_g)\bar{g}} d\bar{y} + \frac{(1 - \alpha_g)\bar{y}}{\bar{y} - (1 - \alpha_g)\bar{g}} d\bar{g}.$$

Rearranging this expression and using Definition 1, one obtains, finally,

$$(A3) \quad \frac{d\bar{y}}{d\bar{g}} = \frac{1 - \alpha_g}{1 + \nu[1 - s_g(1 - \alpha_g)]},$$

as was to be shown.

To establish the second part of Proposition 1, differentiate the long-run GSM with respect to α_g

$$\frac{\partial}{\partial \alpha_g} \frac{\Delta y}{\Delta g} = -\frac{1 + \nu}{(1 + \nu[1 - s_g(1 - \alpha_g)])^2} < 0$$

Thus, $\Delta y / \Delta g$ decreases with α_g , as was to be shown.

PROOF OF PROPOSITION 2:

Part (i): The proof of part (i) of Proposition 2 proceeds as follows. The reduced-form equations (7) and (8) rewrite

$$(A4) \quad \hat{y}_t = \rho \hat{y}_{t-1} + \frac{\alpha}{1 + \alpha \varphi_g} u_t + \frac{1}{1 + \alpha \varphi_g} z_t,$$

$$(A5) \quad \hat{g}_t = \rho \hat{g}_{t-1} + \frac{1}{1 + \alpha \varphi_g} u_t - \frac{\varphi_g}{1 + \alpha \varphi_g} \Delta z_t,$$

where $\Delta z_t = z_t - z_{t-1}$ and

$$\rho \equiv \frac{\alpha \varphi_g}{1 + \alpha \varphi_g}.$$

Provided $\alpha_g < 1$ and $\varphi_g \geq 0$, we have $\rho \in [0, 1)$, so that (A4) and (A5) display second-order stationarity. Accordingly, they can be restated as

$$(A6) \quad \hat{y}_t = \frac{1}{1 + \alpha \varphi_g} \sum_{i=0}^{\infty} \rho^i L^i (\alpha u_t + z_t),$$

$$(A7) \quad \hat{g}_t = \frac{1}{1 + \alpha \varphi_g} \sum_{i=0}^{\infty} \rho^i L^i (u_t - \varphi_g \Delta z_t),$$

where L is the backshift operator. Equation (A7) can be reformulated as

$$(A8) \quad \hat{g}_t = \frac{1}{1 + \alpha \varphi_g} \left[\left(\sum_{i=0}^{\infty} \rho^i L^i \right) u_t - \varphi_g \left(1 + \frac{\rho - 1}{\rho} \sum_{i=1}^{\infty} \rho^i L^i \right) z_t \right].$$

Combining (A6) and (A8), one obtains

$$(A9) \quad E[\hat{y}\hat{g}] = \frac{1}{(1 + \alpha\varphi_g)^2} \left(\frac{\alpha\sigma_u^2}{1 - \rho^2} - \frac{\varphi_g\sigma_z^2}{1 + \rho} \right).$$

Then using the definition of ρ , (A9) rewrites

$$(A10) \quad E[\hat{y}\hat{g}] = \frac{(1 + \alpha\varphi_g)\alpha\sigma_u^2 - \varphi_g\sigma_z^2}{(1 + \alpha\varphi_g)(1 + 2\alpha\varphi_g)}.$$

From (A8), one also obtains

$$(A11) \quad E[\hat{g}^2] = \frac{1}{(1 + \alpha\varphi_g)^2} \left(\frac{\sigma_u^2}{1 - \rho^2} + \frac{2\varphi_g^2\sigma_z^2}{1 + \rho} \right).$$

Then using the definition of ρ , (A11) rewrites

$$(A12) \quad E[\hat{g}^2] = \frac{(1 + \alpha\varphi_g)\sigma_u^2 + 2\varphi_g^2\sigma_z^2}{(1 + \alpha\varphi_g)(1 + 2\alpha\varphi_g)}.$$

Now, combining (A10), (A12), and the definition of $\hat{\alpha}$ yields

$$\hat{\alpha} \equiv \frac{E[\hat{y}\hat{g}]}{E[\hat{g}^2]} = \frac{(1 + \alpha\varphi_g)\alpha\sigma_u^2 - \varphi_g\sigma_z^2}{(1 + \alpha\varphi_g)\sigma_u^2 + 2\varphi_g^2\sigma_z^2},$$

as was to be shown.

Part (ii): The proof of part (ii) of Proposition 2 is straightforward. Form the difference $\hat{\alpha} - \alpha$, which yields

$$(A13) \quad \hat{\alpha} - \alpha = -\frac{\varphi_g\sigma_z^2(1 + 2\alpha\varphi_g)}{(1 + \alpha\varphi_g)\sigma_u^2 + 2\varphi_g^2\sigma_z^2}.$$

This expression is strictly negative whenever $\varphi_g > 0$ and $\sigma_z > 0$, as was to be shown.

Part (iii): To prove the third part of Proposition 2, differentiate (A13) with respect to φ_g . Assuming $\sigma_z > 0$, this yields

$$(A14) \quad \frac{\partial}{\partial \varphi_g} (\hat{\alpha} - \alpha) = -\left(\frac{1}{\omega(1 + \alpha\varphi_g) + 2\varphi_g^2} \right)^2 P(\varphi_g),$$

where we defined $\omega \equiv (\sigma_u/\sigma_z)^2$ and

$$(A15) \quad P(\varphi_g) \equiv 2(\alpha^2\omega - 1)\varphi_g^2 + 4\alpha\omega\varphi_g + \omega.$$

The expression in equation (A14) is defined for $\varphi_g = 0$ whenever $\omega > 0$ (i.e., whenever $\sigma_u > 0$). To simplify the analysis, we henceforth impose this condition. The sign of $\partial(\hat{\alpha} - \alpha)/\partial\varphi_g$ depends on the sign of $P(\varphi_g)$. In turn, the sign of $P(\varphi_g)$ depends on the sign of $\alpha^2\omega - 1$, which calls for the following discussion:

- If $\alpha = \sigma_z/\sigma_u$, then $\alpha^2\omega - 1 = 0$, and thus $P(\varphi_g) = 4\alpha\omega\varphi_g + \omega > 0$ for $\varphi_g \geq 0$. Hence,

$$\forall \varphi_g \geq 0, \quad \frac{\partial}{\partial\varphi_g}(\hat{\alpha} - \alpha) < 0.$$

- Now, if $\alpha > \sigma_z/\sigma_u$, then $\alpha^2\omega - 1 > 0$, and thus $P(\varphi_g)$ admits two negative roots (as can be read from the coefficients). Thus, for $\varphi_g \geq 0$, $P(\varphi_g) > 0$. Hence,

$$\forall \varphi_g \geq 0, \quad \frac{\partial}{\partial\varphi_g}(\hat{\alpha} - \alpha) < 0.$$

- Finally, if $\alpha < \sigma_z/\sigma_u$, then $\alpha^2\omega - 1 < 0$, and thus $P(\varphi_g)$ admits roots of opposite signs (as can be read from the coefficients). Let $\bar{\varphi}_g(\alpha, \sigma_z, \sigma_u)$ define the positive root

$$\bar{\varphi}_g(\alpha, \sigma_z, \sigma_u) \equiv \frac{2\alpha\omega + \sqrt{2\omega(\alpha^2\omega + 1)}}{2(1 - \alpha^2\omega)} > 0.$$

Since the parabola opens downward, $P(\varphi_g) \geq 0$ for $\varphi_g \in [0, \bar{\varphi}_g(\alpha, \sigma_z, \sigma_u)]$ and $P(\varphi_g) < 0$ for $\varphi_g > \bar{\varphi}_g(\alpha, \sigma_z, \sigma_u)$. Thus,

$$\frac{\partial}{\partial\varphi_g}(\hat{\alpha} - \alpha) \leq 0, \quad \forall \varphi_g \in [0, \bar{\varphi}_g(\alpha, \sigma_u, \sigma_z)],$$

$$\frac{\partial}{\partial\varphi_g}(\hat{\alpha} - \alpha) > 0, \quad \forall \varphi_g > \bar{\varphi}_g(\alpha, \sigma_u, \sigma_z).$$

This completes the proof.

PROOF OF PROPOSITION 3:

Let $\mathcal{B}$ denote the impact response bias. It is given by

$$\begin{aligned}
 \mathcal{B} &= \hat{\alpha} - \frac{\alpha}{1 + \alpha\varphi_g} \\
 &= \frac{\alpha(1 + \alpha\varphi_g)\sigma_u^2 - \varphi_g\sigma_z^2}{(1 + \alpha\varphi_g)\sigma_u^2 + 2\varphi_g^2\sigma_z^2} - \frac{\alpha}{1 + \alpha\varphi_g} \\
 &= \frac{\alpha(1 + \alpha\varphi_g)^2\sigma_u^2 - \varphi_g(1 + \alpha\varphi_g)\sigma_z - \alpha(1 + \alpha\varphi_g)\sigma_u^2 - 2\alpha\varphi_g^2\sigma_z^2}{(1 + \alpha\varphi_g)((1 + \alpha\varphi_g)\sigma_u^2 + 2\varphi_g^2\sigma_z^2)} \\
 &= \varphi_g \frac{\alpha^2(1 + \alpha\varphi_g)\sigma_u^2 - (1 + 3\alpha\varphi_g)\sigma_z^2}{(1 + \alpha\varphi_g)((1 + \alpha\varphi_g)\sigma_u^2 + 2\varphi_g^2\sigma_z^2)}.
 \end{aligned}$$

Notice that the sign of $\mathcal{B}$ is equal to the sign of the numerator (up to the parameter φ_g)

$$\alpha^2(1 + \alpha\varphi_g)\sigma_u^2 - (1 + 3\alpha\varphi_g)\sigma_z^2.$$

The threshold value φ_g^* of φ_g , such that $\mathcal{B}$ is zero, is then

$$\varphi_g^* = \frac{\alpha^2(\sigma_u^2/\sigma_z^2) - 1}{\alpha(3 - \alpha^2(\sigma_u^2/\sigma_z^2))}.$$

If $\varphi_g > \varphi_g^*$, the bias $\mathcal{B}$ is negative, and the estimated response on impact always underestimates the true response. We now inspect the sign of φ_g^* , accounting for the restriction $\alpha \in (0, 1]$ (i.e., $\alpha_g \in ((s_g - 1)/s_g, 1]$).

Let α_1 and α_2 (with $\alpha_1 < \alpha_2$) denote the two positive threshold values for α , such that the numerator and the denominator of φ_g^* change sign. These values are given by

$$\alpha_1 = \sigma_z/\sigma_u \quad \text{and} \quad \alpha_2 = \sqrt{3}\sigma_z/\sigma_u.$$

Let us first consider the case when $0 < \alpha_1 < \alpha_2 < 1$. If $\alpha < \alpha_1$, the numerator is negative while the denominator is positive. Then, φ_g^* is negative. If $\alpha_1 < \alpha < \alpha_2$, the numerator and the denominator are positive, and thus $\varphi_g^* > 0$. Finally, if $\alpha > \alpha_2$, the numerator is positive while the denominator is negative. Then, φ_g^* is negative. When $\varphi_g^* < 0$, any degree of countercyclicality in the endogenous component of policy satisfies $\varphi_g > 0 > \varphi_g^*$ and thus the bias $\mathcal{B}$ is always negative, that is, the estimated response underestimates the true one. When $\varphi_g^* > 0$, the endogenous component of government policy must be sufficiently countercyclical in order to get an downward-biased response.

Consider now the case when $0 < \alpha_1 < 1 < \alpha_2$. When $\alpha < \alpha_1$, we get $\varphi_g^* < 0$, and the bias $\mathcal{B}$ is negative. When $\alpha > \alpha_1$, we get $\varphi_g^* > 0$, and φ_g must exceed $\varphi_g^* < 0$ in order to obtain a negative bias.

Finally, consider the configuration $1 < \alpha_1 < \alpha_2$. It follows that $\varphi_g^* < 0$. The bias $\mathcal{B}$ is then negative and the estimated response underestimates the true one.

PROOF OF PROPOSITION 4:

To prove the first part of Proposition 4, we start by characterizing the deterministic steady state. The latter is defined in terms of detrended variables. To be more specific, we induce stationarity according to the following formulas:

$$c_t^s \equiv c_t e^{-z_t}, \quad x_t^s \equiv x_t e^{-z_t}, \quad g_t^s \equiv g_t e^{-z_t}, \quad y_t^s \equiv y_t e^{-z_t}, \quad k_{t+1}^s \equiv k_{t+1} e^{-z_t}.$$

Using these definitions and the equilibrium conditions, the model steady state is solution to the system of equations

$$(A16) \quad \bar{c}^s + \bar{x}^s + \bar{g}^s = \bar{y}^s,$$

$$(A17) \quad \bar{y}^s = \left(\frac{\bar{k}^s}{\gamma_z} \right)^\theta \bar{n}^{1-\theta},$$

$$(A18) \quad \frac{1}{\bar{c}^s + \alpha_g \bar{g}^s} (1 - \theta) \bar{y}^s = \eta (1 - \beta \phi) \bar{n}^{(1+\nu)(1-\phi)},$$

$$(A19) \quad 1 = \frac{\beta}{\gamma_z} \left(1 - \delta + \frac{\theta \gamma_z \bar{y}^s}{\bar{k}^s} \right),$$

$$(A20) \quad \bar{x}^s = \left(1 - \frac{1 - \delta}{\gamma_z} \right) \bar{k}^s.$$

From (A19) and (A20), one can solve for the capital-output and investment-output ratios

$$\frac{\bar{y}^s}{\bar{k}^s} = \frac{\gamma_z - \beta(1 - \delta)}{\beta \theta \gamma_z},$$

$$s_x = \frac{\bar{x}^s}{\bar{y}^s} = \frac{\beta \theta (\gamma_z - 1 + \delta)}{\gamma_z - \beta(1 - \delta)}.$$

It follows from these relations that s_x and $\bar{y}^s/\bar{k}^s$ do not depend on $\bar{g}^s$. Thus, the ratio $\bar{y}^s/\bar{n}$ does not either depend on $\bar{g}^s$.

Since $\bar{x}^s/\bar{y}^s$ does not depend on $\bar{g}^s$, it must be the case that

$$\frac{d\bar{x}^s}{d\bar{g}^s} = s_x \frac{d\bar{y}^s}{d\bar{g}^s}.$$

Now, differentiating equation (A16) with respect to $\bar{g}^s$ yields

$$(A21) \quad \frac{d\bar{c}^s}{d\bar{g}^s} = (1 - s_x) \frac{d\bar{y}^s}{d\bar{g}^s} - 1.$$

This establishes part (ii) of the proposition.

Differentiating equation (A18), one obtains

$$(s_c + \alpha_g s_g) \frac{d\bar{y}^s}{d\bar{g}^s} = (1 + \nu)(1 - \phi)(s_c + \alpha_g s_g) \frac{\bar{y}^s}{\bar{n}} \frac{d\bar{n}}{d\bar{g}^s} + \frac{\partial \bar{c}^s}{\partial \bar{g}^s} + \alpha_g,$$

where

$$s_c \equiv \frac{\bar{c}^s}{\bar{y}^s}.$$

Now, since the ratio $\bar{y}^s/\bar{n}$ does not depend on $\bar{g}^s$, it must be the case that

$$\frac{1}{\bar{n}} \frac{d\bar{n}}{d\bar{g}^s} = \frac{1}{\bar{y}^s} \frac{d\bar{y}^s}{d\bar{g}^s},$$

so that the previous equation rewrites

$$(s_c + \alpha_g s_g)[1 - (1 + \nu)(1 - \phi)] \frac{d\bar{y}^s}{d\bar{g}^s} = \frac{d\bar{c}^s}{d\bar{g}^s} + \alpha_g.$$

Using equation (A21) and Definition 1, one arrives at

$$\frac{\Delta y}{\Delta g} = \frac{1 - \alpha_g}{1 - s_x + \mu[1 - s_x - s_g(1 - \alpha_g)]},$$

where we made use of the identity $s_c = 1 - s_x - s_g$.

To prove the last part of Proposition 4, differentiate the long-run GSM with respect to α_g , which implies

$$\frac{\partial}{\partial \alpha_g} \frac{\Delta y}{\Delta g} = - \frac{(1 - s_x)(1 + \mu)}{(1 - s_x + \mu[1 - s_x - s_g(1 - \alpha_g)])^2} < 0.$$

This completes the proof.

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